

Eduardo Venegas Hernandez

Data & AI Engineer | Munich, Germany
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EDUCATION

MSc. Data Science and Artificial Intelligence Universität des Saarlandes (Saarbrücken, Germany) 10/2024 - 09/2026

- Cousing: *Statistics, ML Cybersecurity, NLP, Machine Translation, Image Processing & Computer Vision.*
- **Thesis (Seminar Grade 1.0):** Evaluating **Multi-Turn Agentic RAG Systems** (In collaboration with **BMW & DFKI**).

BSc. Computer Science Tecnológico de Monterrey GDL (Guadalajara, Mexico) 08/2019 - 12/2023

- *ML, Time series, Algorithms & Optimization, Cloud Computing & Distributed, Vehicle User Interfaces, PLC.*

Computer Science Exchange Student Universität Ulm (Ulm, Germany) 10/2022 - 09/2023

- Full scholarship recipient from **DAAD**. *Computer Vision II, Autonomous Driving and Assistance Systems, Statistics.*

Technologist Degree in Automation and Robotics CETI Colomos (Guadalajara, Mexico) 08/2015 - 06/2019

- *Control Laboratory (I & II), Industrial Robotics, PLCs, Digital Electronics, Controllers Laboratory, Sensor Systems*

SKILLS

AI & Data: PyTorch, LangGraph, LangChain, RAG, Knowledge Graphs, MCP, Hugging Face, TensorFlow, Vector Databases, SQL

Software & Frameworks: Python, C/C++, TypeScript/JavaScript, FastAPI, Flask, Node.js, React, Next.js, Angular, Tauri

Systems & DevOps: Linux, Docker, Kubernetes, CI/CD, AWS, Azure, Git, Grafana, MQTT, PLCs

WORK EXPERIENCE

Data & AI Engineer Masterarbeit BMW & DFKI (Munich, Germany) 03/2026 - 09/2026

- Topic: Evaluating **Multi-Turn Agentic RAG Systems: A Synthetic Benchmark Generation** Approach.
- Architected an evaluation framework for **multi-turn tool-using Agentic RAG** systems enabling precise failure localization.
- Built a multi-turn synthetic benchmark pipeline that combines deterministic **task planning** with diverse user **personas**.
- Engineered an **unsupervised audit framework** for a **RAG system** across **4M unannotated** corporate records.

AI Research Engineer Werkstudent August-Wilhelm Scheer Institute (Saarbrücken, Germany) 10/2024 - 02/2026

- Designed and deployed a **KG-RAG** over unstructured technical documentation using **LangChain** and **HuggingFace**.
- Developed an **Autoencoder model** for real-time **anomaly detection** across **distributed devices** via **Federated Learning**.

Big Data Platform Engineer Oracle (Guadalajara, Mexico) 11/2023 - 10/2024

- Engineered **CI/CD testing pipelines** and deployment tools of **big data clusters** on **Oracle Cloud Infrastructure (OCI)**.
- Maintained **automation frameworks** within the **Big Data Service** team under **Oracle Analytics**.

Data Science & Analytics Intern BMW Group (Munich, Germany) 03/2023 - 09/2023

- Developed a high-accuracy **anomaly detection algorithm** specialized in identifying external **battery replacements**.
- Engineered scalable **time-series data pipelines** using **PySpark** to process raw **vehicle telemetry** and **cloud logs**.
- Optimized **data flows**, resolving missing **database values** to increase **data reliability** and a 62% increase in **service utilization**.
- Allowed adjusting **predictive model parameters** and receive real-time **system performance outputs**.

AI & DevOps Engineer Xira (Mexico City, Mexico) 01/2022 - 04/2022

- Managed over **distributed server infrastructure**, established **service monitoring** and enhanced engineering processes.
- Built an **API management platform**, integrating payments and subscription systems using **NextJS**, **Stripe** and **Docker**.

Production Engineer Fellow Major League Hacking & Meta (New York, USA) 06/2021 - 09/2021

- Selected internationally among 120 out of 20,000 applicants. Implemented **monitoring** with **Prometheus** and **Grafana**.
- Automated **CI/CD workflow** for efficient deployment to a production server using **Docker** in an **AWS EC2 instance**.

Machine Learning & Full Stack Engineer Jiit Technologies (Guadalajara, Mexico) 10/2020 - 06/2021

- Designed a custom **sales forecasting** and **data analytics API pipeline** using **Flask**, **SciPy** and **Prophet**.
- Built an **ERP platform** to handle multi-module operations (inventory, logistics, and finance) utilizing **Angular**.

PROJECTS

VectorDrive-Route: Autonomous BEV Trajectory Planning Github

- Built a **PyTorch pipeline** using **nuScenes**, **ResNet** and **depth lifting** to map multi-camera feeds into a **Bird's-Eye-View grid**.
- Designed a temporal **motion planning head** with **drivable-area** and **occupancy decoders** to generate safe paths.

Daten-Mapping-Framework für die AAS DOI: 10.17560/atp.v68i5.2836

- Developed a mapping framework from **schemas** to **Asset Administration Shell** to achieve interoperability in **Industry 4.0**.
- Designed a dual alignment strategy separating standard **relational tables** from **hierarchical assets**.

See Your World Github

- Crafted a hackathon award-winning app for the visually impaired, supporting 11 languages.
- Engineered **Image Captioning DL model** with **TensorFlow**, deployed using **Flask**, **Docker**, and **Google Cloud VM**.

CERTIFICATES

- **Mathematics for Data Science Specialization** | HSE - 2021
- **Data Science Professional Certificate** | Harvard - 2020
- **Python Data Science Professional Certificate** | IBM - 2020
- **Deep Learning Professional Certificate** | IBM - 2020

LANGUAGES

- **Spanish:** Native
- **English:** C1 (TOEFL iBT)
- **German:** C1 (Goethe-Institut)